

parties contract with each other to borrow or lend.³⁵ All other risky securities—stocks, corporate bonds, and so on—are in *positive net supply* and represent real wealth. Thus, for corporate bonds, I recommend starting with the standard factor investing procedure.³⁶

This same concept applies to all government debt, not only short-term T-bills. *Ricardian equivalence* is the name given to the principle that any liability of a government has to be repaid at some point (“tax now or tax later”), and in simple economic models without frictions, government debt is not net wealth: if we wrote down an aggregate balance sheet for the economy across generations, we would also see zero net supply.

The profession has moved beyond these simple cases and shown that government debt can be net wealth and fulfills many purposes other than borrowing or lending.³⁷ The simple economic models are useful to show that *the baseline case for investing in sovereign safe assets should not be market weights*. Market weights for sovereign debt are not very meaningful. To determine optimal holdings of safe assets, we should understand the roles that safe assets play and why governments issue debt, how much, and how often they default.

6.2. FACTORS IN SAFE ASSETS

Government debt, like any other asset, is simply a bundle of risk factors. The problem is that so many risk factors, some acting at cross purposes, are at work in safe assets. Look at what the International Monetary Fund (IMF) says:³⁸

Safe assets are used as a reliable store of value and aid capital preservation in portfolio construction. They are a key source of liquid, stable collateral in private and central bank repurchase (repo) agreements and in derivatives markets, acting as the lubricant or substitute of trust in financial transactions. As key components of prudential regulation, safe assets provide banks with a mechanism for enhancing their capital

³⁵ A major paper by Cox, Ingersoll, and Ross (1985) shows there are few restrictions on the risk-free interest rate process in equilibrium. They assume all safe bonds are in zero net supply.

³⁶ Aren't corporate bonds similar in that for every company that issues (sells) bonds there is a buyer? Why are corporate bonds net wealth? Miller and Modigliani (1958, 1961) tell us that in frictionless markets, there is no difference between corporate bonds and equities. We know this world is far from reality, but the main point that corporate bonds and equity are both claims on a risky corporation that generates profitable investment opportunities, on average, remains the same. This is not true for risk-free government debt.

³⁷ David Ricardo, after whom Ricardian equivalence is named, was one of the giants who established economics as a field in the early 1800s. The seminal reference is Barro (1974), who overcomes Ricardian equivalence by assuming distortionary taxes. Lucas and Stokey (1983) is another standard reference. A summary article is Ricciuti (2003).

³⁸ IMF, Global Financial Stability Report, 2012.